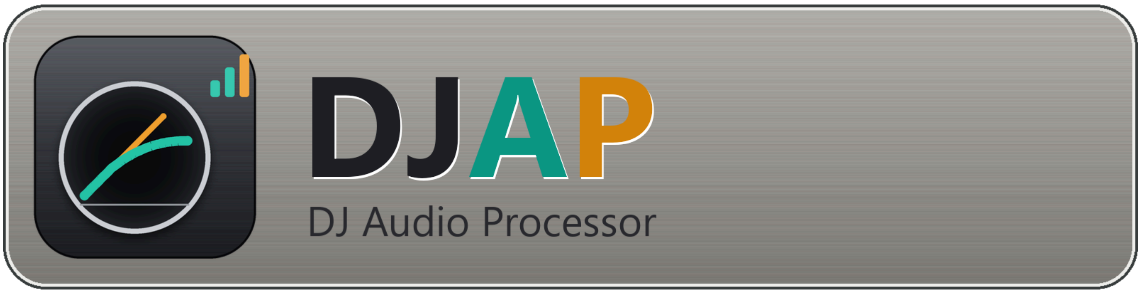
DJAP
DJ Audio Processor

DJAP user manual

Installing, connecting and using DJAP: every window, control and setting

Version 0.1.176

https://github.com/hghaken/DJAP

Free software under the GNU General Public License v3

DJAP (DJ Audio Processor) is a loudness and dynamics processor for Windows. It sits between a sound source and the amplifier or recorder and makes every song come out at the same loudness, with a controlled sound and peaks that never pass a set ceiling. This manual explains how to install it, connect it, and use every window, control and setting. How the processing works inside is described in the separate technical manual (`DJAP-Technical-Manual.pdf` in the installation folder; `docs/technical.md` on https://github.com/hghaken/DJAP).

This manual describes DJAP from version 0.1.96 on (the first version with updates from GitHub); the plug-in for DJ software (chapter 9) came with the version after 0.1.100.

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1. What DJAP does

DJAP has two independent stereo channels, **deck A** and **deck B**. In a DJ set-up each deck of the DJ software or each channel of the mixer is sent through its own DJAP deck, so both decks are processed separately and come out at the same loudness. For other use - for example everything Windows plays - DJAP can also run as **one channel**: one stereo processor. The **plug-in** (chapter 9) brings the same processing into DJ software itself, one deck per plug-in.

Each deck runs the same chain of processing stages:

Step	What it does for you
Leveller	A slow, automatic hand on the volume: quiet and loud songs come out at the same loudness. It does not follow the music within a song and holds still during fades and pauses; a quiet part that stays (a soft verse, a breakdown) comes up to the target after a few seconds.
Colour	The tone of the music: bass, low-mid, presence and treble, an automatic tone correction per song, a sub-bass generator and a contour for quiet listening.
Compressor	Evens out the loud parts within a song.
Density	A multiband stage (1 to 7 frequency bands): a band that sticks out is turned down, one that falls away is lifted - the full, dense "radio" sound.
Bass enhancer	Overtones of the bass, so the bass is heard on small speakers.
Exciter	Overtones in the highs: shine and air.
De-esser	Takes the sharpness off "s" and "t" sounds.
Stereo width	A wider or narrower stereo image; the bass can be made mono.
Soft clipper	Rounds off the highest peaks, without delay.
Limiter	The safety net: the output never passes the ceiling (true peak, default -1 dBTP).
Output gain	A last trim, only downwards.

What each step does and how much is set by the deck's **loudness preset** (the loudness and dynamics) and its **colour** (the tone). DJAP ships with 8 loudness presets and 9 colours; every value can be changed and saved under your own name.

Things to know up front:

- **Low latency.** The only latency DJAP adds is the limiter's look-ahead: 1.5 ms by default (0, 1.5, 2, 3 or 5 ms), the same for both decks so they stay in time with each other when mixed.
- **Bypass is exact.** With a deck on Bypass its input goes out unchanged, bit for bit, delayed by the same look-ahead - so switching between processed and unprocessed never shifts the sound in time.
- **No clicks.** Every change - a loudness preset, a colour, a slider, a stage switched on or off - glides in over a few milliseconds. Starting and stopping the audio fade in and out.
- **Everything is saved at once.** There is no Save button: every change is stored in the settings file.

2. Installing, updating and removing

2.1 What you need

- Windows 10 or 11, 64-bit.
- For two decks: an audio interface with an **ASIO** driver and at least 4 inputs and 4 outputs (two stereo pairs each way).
- For one channel (Windows' own sound): any sound card. To run Windows' sound through DJAP you need a virtual audio cable, such as VAC (Virtual Audio Cable; the free Lite version is enough) - see 3.2.

No other software is needed: the programs carry their own runtime.

2.2 Installing

Run `DJAP-Setup-<version>.exe`. The setup is not signed, so Windows SmartScreen may warn before the first run ("More info", then "Run anyway").

The setup checks the computer first and shows what it found on the last page before installing:

- **Windows components** the programs use. If one is missing the setup stops and says which.
- **Media Foundation.** The "N" editions of Windows lack it. DJAP itself runs without it, but the command-line program DJAPCmd needs it to read MP3, M4A, WMA and FLAC files. The setup offers to add Microsoft's Media Feature Pack from Windows Update.
- **VAC.** Only needed for Windows' own sound with one channel. When it is missing the setup explains what it is for and can open its download page at the end. The setup never installs it.
- **ASIO drivers** found on the computer.

Options on the way:

- **Select Components**: *DJAP* (the program) and the *Plug-in for DJ software* (chapter 9); both are chosen by default, each can be left out.
- **Create a desktop shortcut** (off by default).
- **Start DJAP in the notification area when Windows starts, with the audio devices used last** (on by default). This can be changed later in the Setup window.

DJAP is installed in `C:\Program Files\DJAP`. Installing a newer version over an older one keeps all settings and own loudness presets. If DJAP is running, the setup asks to close it first.

2.3 The first start

At the very first start no audio device has been chosen yet, so the **Setup window** opens by itself. Choose the processing (two decks or one channel), the audio API and the devices, then press **Start audio** - see chapter 3 and chapter 5.

2.4 The notification area

DJAP lives in the notification area (the icons next to the clock). Closing or minimising the main window hides it there; **the audio goes on**. The first time this happens a message says so.

- **Click** the icon to show the window again.
- **Right-click** the icon for its menu: *Open DJAP*, *Setup...*, *About DJAP...*, *Shutdown*.
- The icon's tooltip shows whether the audio runs and each deck's loudness preset and colour (or "bypass").

DJAP really ends only with **Shutdown** (from the icon's menu or the main window's right-click menu), when Windows shuts down, or when a setup closes it. Before it stops, the audio fades out over 10 ms, so there is no click.

Only one DJAP runs at a time: starting it a second time brings the running one to the front.

2.5 Starting with Windows

With "Start DJAP when Windows starts" on (Setup window, or the setup's option), DJAP starts hidden in the notification area when you log on, and starts the audio with the devices of the last session if the audio was running when it was last closed. Devices that are not ready yet right after logging on are tried again every 5 seconds for 2 minutes. If you stopped the audio before closing DJAP, it stays stopped at the next start.

Windows' own list of startup apps (Settings > Apps > Startup) can switch it off too; DJAP shows that as off.

2.6 Updating

DJAP checks once a day whether a newer version has been published and tells you; installing it takes one click. See chapter 11.

2.7 Removing

Use Windows' Settings > Apps > Installed apps > DJAP > Uninstall. The uninstaller removes the programs, the shortcuts and the "start with Windows" entry. It then asks whether to delete your settings and own loudness presets too (`%APPDATA%\DJAP`); answer No to keep them for a later installation.

3. Connecting DJAP

3.1 DJ set-up with an ASIO interface (two decks)

DJAP needs the two decks as separate stereo signals and gives back two processed stereo signals:

```
deck A source -> interface inputs 1-2 -> DJAP deck A -> interface outputs 1-2 -> mixer / amplifier
deck B source -> interface inputs 3-4 -> DJAP deck B -> interface outputs 3-4 -> mixer / amplifier
```

The channel pairs can be chosen freely in the Setup window (any input pair and any output pair per deck, as long as the two decks use different output pairs).

Where to take the deck signals from:

- **Before the channel fader and crossfader (pre-fader)** is best: the leveller then settles on a song while you pre-listen, and your fader moves are not "corrected".
- **After the fader** also works: the leveller's hold rule keeps fades and cuts from being pumped up, but it is never as clean as pre-fader.

EQ kills and filter sweeps look like level changes to the leveller. Deep cuts are caught by its hold rule; with lighter EQ moves the leveller may make up part of the cut over a few seconds. A slower leveller (loudness window, Leveller > Speed) makes that less noticeable.

An ASIO driver usually serves one program at a time. If your DJ software uses the same interface through ASIO, DJAP may not be able to open it, or opening it may take the interface away from the DJ software. Use a second interface, or route the DJ software's deck outputs to the inputs DJAP reads through the interface's own mixer software. DJAP always asks before it opens a driver.

3.2 Windows' own sound (one channel)

To process everything Windows plays (browser, music player, video):

1. Install a virtual audio cable (VAC). It adds a playback device ("Line 1") and a recording device of the same name connected to it.
2. Make the cable's playback device the **default playback device** in Windows' sound settings. All Windows sound now goes into the cable.
3. In DJAP's Setup window choose **Processing: One channel**, **Audio API: WASAPI**, as **Input** the cable's recording device and as **Output** your speakers or headphones.
4. Press **Start audio**.

With one channel, the outputs follow the Windows volume by default (the keyboard's volume keys and the taskbar slider), after the processing - see 5.2. Without this, the volume keys would turn down the cable and the leveller would turn it back up.

You can also keep an ASIO interface as the output and take the input from a WASAPI device: choose **Audio API: ASIO** and **Input from: WASAPI devices** (see 5.1, "Deck inputs"). The two devices run on separate clocks; DJAP corrects the drift between them at the cost of a buffer (about 15-20 ms with a 10 ms WASAPI input; see chapter 10).

3.3 Which audio API?

API	Latency	Use it for
ASIO	lowest	Audio interfaces with an ASIO driver; the only API where both decks run in one synchronous stream. Best for DJs.
WASAPI shared	low to medium	Windows' own audio system, shared with other programs. Each deck has its own input and output device.

API	Latency	Use it for
WASAPI exclusive	low	As shared, but DJAP takes the devices for itself while the audio runs (DJAP asks first).
DirectSound, **MME**	higher	Older interfaces; for devices without a good driver.

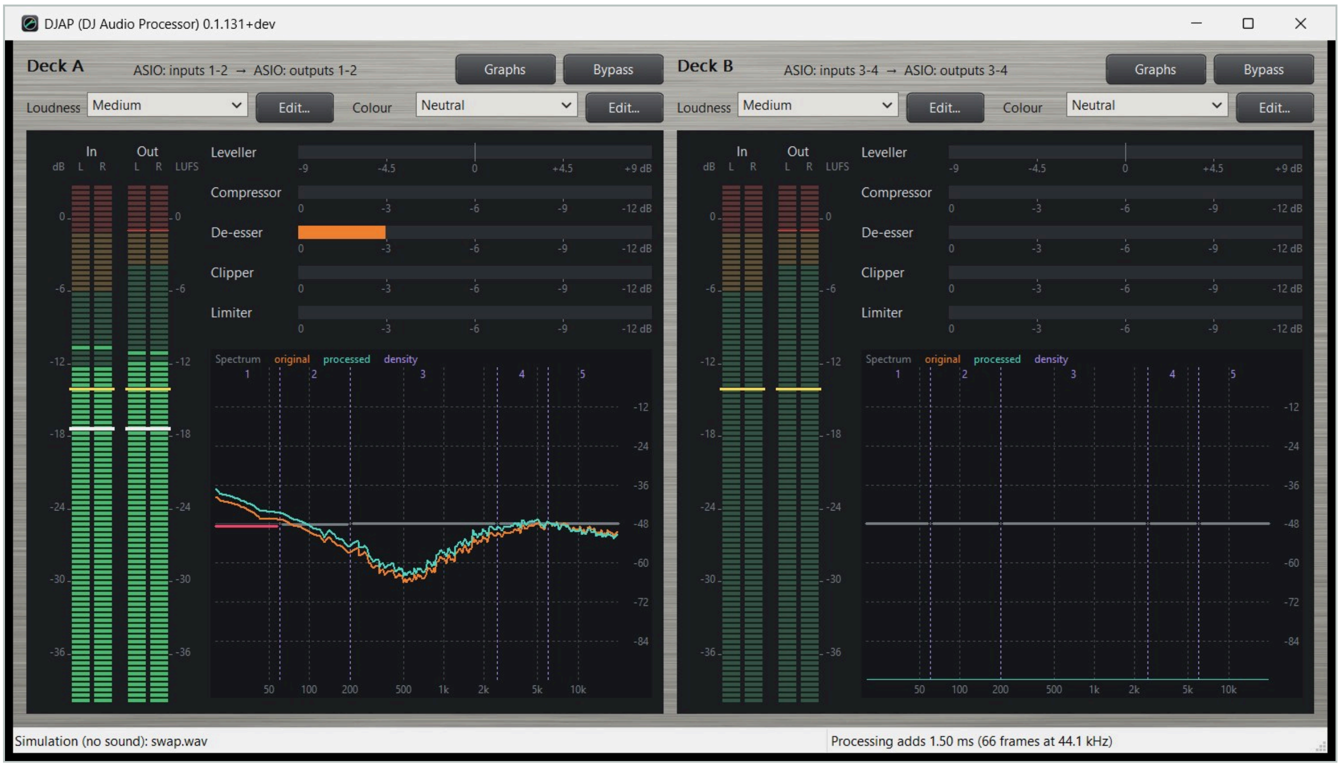
With every API the processing itself adds only the limiter look-ahead; the rest of the latency is the devices' buffers (chapter 10).

With WASAPI, DirectSound and MME a deck's input and output device usually run on clocks of their own (a virtual cable and a USB headset, a recording device and the PC's speakers), and two clocks always drift apart a little. With **Drift correction** (Setup window) on *Auto*, the default, DJAP reads the input a little faster or slower - at most 0.2 %, inaudible - so they never drift apart, at the cost of a buffer of about 8 ms (128-frame periods) to 23 ms (10 ms periods), shown while running. With WASAPI an input and output of the same device (one clock, for example the inputs and outputs of one USB interface) are recognised and run without the buffer, as one stream - 5 to 13 ms less. Should such a device still give a short glitch now and then, choose *Always*: then it is corrected too. *Off*: the input always goes straight to the output with no buffer, and devices on different clocks give a short glitch now and then.

3.4 Test safely

Opening a driver or a device in exclusive mode takes it from other programs, and outputs you think are silent may still be routed to speakers. For a first test, use outputs nobody listens to, start with a low volume on the amplifier, and watch the meters before you turn up.

4. The main window


DJAP (DJ Audio Processor) 0.1.131+dev
Deck A
ASIO: inputs 1-2 → ASIO: outputs 1-2
Graphs
Bypass
Loudness
Medium
Edit...
Colour
Neutral
Edit...
Deck B
ASIO: inputs 3-4 → ASIO: outputs 3-4
In
Out
dB L R
L R LUFS
Leveller
Compressor
De-esser
Clipper
Limiter
-9
-4.5
0
+4.5
+9 dB
0
-3
-6
-9
-12 dB
Spectrum
original
processed
density
1
2
3
4
5
-12
-24
-36
-48
-60
-72
-84
0
-6
-12
-18
-24
-30
-36
50
100
200
500
1k
2k
5k
10k

Simulation (no sound): swap.wav

Processing adds 1.50 ms (66 frames at 44.1 kHz)

The main window shows one strip per deck (one strip, "Channel", over the whole width with one channel). The window has no menu bar: **right-click anywhere in it** for *Setup...*, *About DJAP...* (the version, the settings file in use and the licences) and *Shutdown*. Every control has a tooltip that explains it when the mouse rests on it. The window can be made as large as you like; it cannot be made smaller than the size that still shows every meter row and a readable spectrum.

4.1 The deck strip

- **Title row.** The deck's name and its route on two lines, for example `In: ASIO: inputs 1-2` above `Out: ASIO: outputs 1-2`, or with other APIs the device names (a name too long for the window ends in "..."; the Setup window shows it whole). With linked presets deck B shows "linked to deck A".
- **Graphs** opens the analysis window (8.1).
- **Bypass.** Lit (green): the deck's input goes out unchanged, delayed by the look-ahead. Not lit: the deck is processed. Saved per deck.
- **Loudness** and its **Edit...** button: the deck's loudness preset (the loudness and dynamics) and the loudness window (chapter 6). The list runs from the quietest output to the loudest; your own loudness presets are sorted in.
- **Colour** and its **Edit...** button: the deck's colour (the tone) and the colour window (chapter 7).

A new loudness preset or colour is heard at once, without a click.

4.2 The bass settings

Four settings for the most-used bass values, in the spectrum window (8.2: a click on the meter panel's spectrum opens it) as sliders - the bass level and frequency also as the blue point in its graph - and in the plug-in's window (9). They change the loudness preset or the colour itself, so every deck using that loudness preset or colour changes with it. A change is heard while dragging and saved when the slider is let go. Arrow keys and Page Up / Page Down step the slider that has the focus.

Slider	Range	What it does	Belongs to
Bass level	-12 .. +12 dB	More or less bass below the bass frequency (the colour's bass shelf). The leveller keeps the loudness, so this changes the balance, not the volume. 0 dB = off.	colour
Bass enhancer	0 .. 100 %	Strength of the bass overtones, so the bass is heard on small speakers. 0 % switches the enhancer off; more switches it on.	loudness preset
Bass width	0 .. 100 %	Stereo width of the bass: 0 % = bass in mono (tighter in a club, safe on mono systems), 100 % = as recorded. Moving it switches the stereo-width stage on.	loudness preset
Bass frequency	40 .. 250 Hz	Where "bass" ends for the three sliders above: the colour's bass corner and, within their own ranges, the enhancer's frequency (60-160 Hz) and the width stage's bass split. Each can be set separately in the loudness preset and colour windows.	both

4.3 The meter panel

The dark panel under the Loudness and Colour lists shows what the deck is doing.

LED bars, left. *In L / R* is the input level (sample peak in dBFS, including the loudness preset's input gain); *Out L / R* is the output level (true peak in dBTP). Lit segments show the level; the zone colours show amber and red near the top (input: amber from -6, red from -1 dBFS; output: amber in the last 3 dB below the ceiling, red above it). A single LED holds the peak for 1.5 s. Across each pair a **white** line shows the momentary loudness and a **yellow** line the target loudness. The scale on the left is in dB, the one on the right in LUFS; they share one axis, so the white line can be compared with the bars and with the target directly.

Stage rows, right. What each stage does now:

- **Leveller** - its gain, up (right of 0) or down (left), over its range.
- **Compressor**, **De-esser**, **Clipper**, **Limiter** - gain reduction, 0 to 12 dB.

A stage that is switched off in the loudness preset is shown greyed out.

Spectrum. The frequency content of the input (**original**, orange) and the output (**processed**, turquoise), 20 Hz to 20 kHz. The density stage is drawn in it: its bands (dotted lines at the splits, band numbers at the top) and each band's gain now as a line over the band's range (purple, like the bands), up or down by up to 12 dB. The **colour** (yellow) is the deck's colour as one EQ curve on the same scale, the auto tone's corrections of the moment included (its points and per-section curves are in the spectrum window, 8.2); in Bypass it is not drawn.

Clicking in the panel opens the setting behind it: a stage row opens the loudness window on that stage's tab, the In bars open the General tab (input gain, target), the Out bars the Limiter tab (ceiling), the spectrum the spectrum window (8.2).

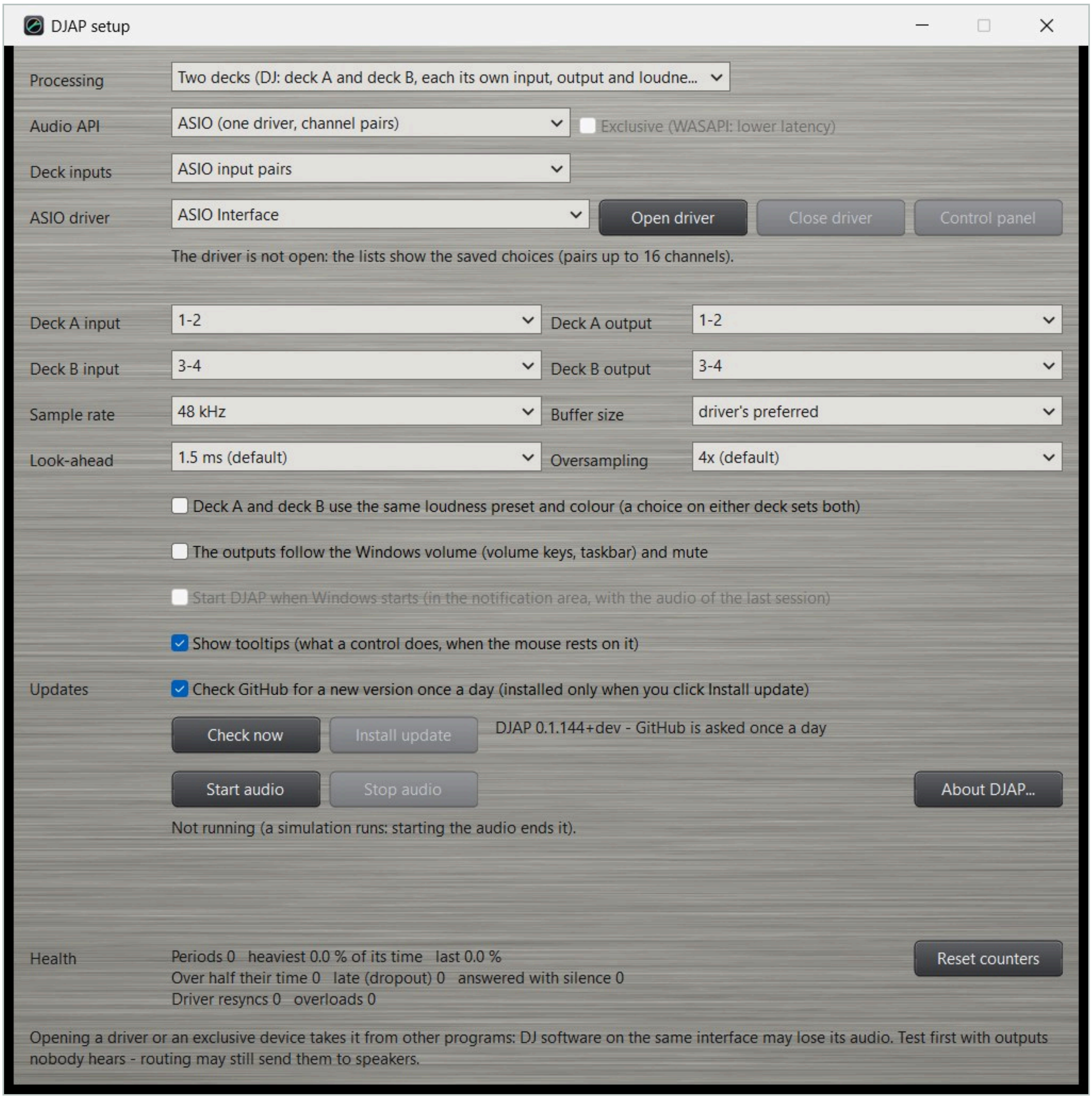
4.4 The status bar

The Windows volume (or "muted") and DJAP's CPU use, as a share of the whole computer as Task Manager shows it (two decks take a few percent of one processor core, so well under 1 % on a computer with many cores). After them, only when there is one: an error, a start that is being retried, a settings file that could not be saved, a new version. The audio's state - devices, sample rate, buffer size, latency, drift, the health counters - is in the Setup window (chapter 5).

4.5 Linked presets

With "Deck A and deck B use the same loudness preset and colour" on (Setup window), the main window shows one set of controls - deck A's loudness preset, colour, Edit buttons and Bypass - and they count for both decks (the spectrum window's bass sliders too). Deck B's strip shows only its route and its meters.

5. The Setup window



Processing: Two decks (DJ: deck A and deck B, each its own input, output and loudne...

Audio API: ASIO (one driver, channel pairs)   ☐ Exclusive (WASAPI: lower latency)

Deck inputs: ASIO input pairs

ASIO driver: ASIO Interface

The driver is not open: the lists show the saved choices (pairs up to 16 channels).

Deck A input: 1-2   Deck A output: 1-2

Deck B input: 3-4   Deck B output: 3-4

Sample rate: 48 kHz   Buffer size: driver's preferred

Look-ahead: 1.5 ms (default)   Oversampling: 4x (default)

☐ Deck A and deck B use the same loudness preset and colour (a choice on either deck sets both)

☐ The outputs follow the Windows volume (volume keys, taskbar) and mute

☐ Start DJAP when Windows starts (in the notification area, with the audio of the last session)

☑ Show tooltips (what a control does, when the mouse rests on it)

Updates: ☑ Check GitHub for a new version once a day (installed only when you click Install update)

DJAP 0.1.144+dev - GitHub is asked once a day

Not running (a simulation runs: starting the audio ends it).

Health:
Periods 0   heaviest 0.0 % of its time   last 0.0 %
Over half their time 0   late (dropout) 0   answered with silence 0
Driver resyncs 0   overloads 0

Opening a driver or an exclusive device takes it from other programs: DJ software on the same interface may lose its audio. Test first with outputs nobody hears - routing may still send them to speakers.

Right-click the main window (or the notification-area icon) and choose **Setup...**. Every choice is saved at once. While the audio runs, a new device, channel pair, sample rate, buffer size or input API is applied at once by starting the audio again (a short fade-out and fade-in).

5.1 Audio

Row	What it does
Processing	*Two decks* (deck A and deck B, each its own input, output and loudness preset) or *One channel* (one stereo processor, for example for what Windows plays).
Audio API	ASIO, WASAPI, DirectSound or MME (see 3.3). Choosing another API stops the audio. **Exclusive** (WASAPI only): lower latency, but the devices are taken from other programs while the audio runs.

Row	What it does
Drift correction (WASAPI, DirectSound, MME)	*Auto* (default): each deck's input is read a little faster or slower so that an input and output device on different clocks never drift apart; adds a buffer of about 8-23 ms (smaller with a smaller buffer size). WASAPI leaves an input and output of the same device as one stream without it. *Always*: also those. *Off*: no buffer, but a glitch now and then unless input and output are the same device (see 3.3). Changing it while the audio runs starts the audio again.
Deck inputs / Input from (ASIO only)	Where the decks' inputs come from: *ASIO input pairs*, or devices of WASAPI, DirectSound or MME (drift-corrected; adds a buffer of about 15-20 ms with a 10 ms input). The outputs stay ASIO.
Input period (ASIO with WASAPI inputs; in the place of Drift correction)	How often Windows hands over the input's sound. *Windows' default (10 ms)*, the default. *Short, as the ASIO buffer*: as often as the ASIO buffer, when the device has Windows' low-latency mode - the input's 10 ms and part of the drift buffer go, up to about 14 ms less latency. A device without that mode keeps 10 ms, and the Health section then says "the input device has no periods this short". Try it while you listen and watch the underruns; changing it while the audio runs starts the audio again.
ASIO driver	The installed ASIO drivers. **Open driver** opens it (DJAP asks first: it may take the interface from other programs); **Close driver** releases it; **Control panel** opens the driver's own settings (buffer size, clock source). Under it: the driver's state.
Deck A / B input, **output**	ASIO: a channel pair per deck, with the driver's channel names once it is open. Other APIs: an input and an output device per deck; *(none)* = the deck does not run. A saved device that is not connected shows "(not found)".
Sample rate	44.1, 48, 88.2 or 96 kHz. ASIO uses this rate if the driver accepts it, else the interface's own rate. Other APIs convert when a device runs at another rate.
Buffer size	ASIO: the driver's preferred size, or a size it allows. Other APIs: the period asked of Windows. Smaller = less latency, more risk of dropouts (watch Health).
Look-ahead	The limiter's look-ahead: 0, 1.5 (default), 2, 3 or 5 ms, the same for both decks. Longer handles peaks more gently. Every choice above 0 limits the true peak; at 0 ms the limiter holds only the sample peak (true peaks can pass the ceiling) and adds no delay. Changed at once with a short fade. (The former 0.5 and 1.0 ms were too short for the true-peak detector and are now 1.5 ms.)
Oversampling	2x, 4x (default) or 8x, both decks: how many times the sample rate the soft clipper and the exciter work at. Their overtones above the audio band would fold back as harsh tones that do not belong to the music; 2x leaves a rest on hard-driven high tones (about 40 dB under a 15 kHz tone), 4x and 8x take it away (60 to more than 100 dB under) at a little more CPU. Adds no latency. Changed with a short fade of those two stages.

5.2 Options

- **Deck A and deck B use the same loudness preset and colour** - linked presets (4.5).
- **The outputs follow the Windows volume (volume keys, taskbar) and mute.** The volume of Windows' default playback device turns DJAP's outputs down after the processing (never up: the ceiling holds). The meters show the level before this volume. If the default playback device is the virtual cable DJAP reads, and Windows turns that cable down in software, DJAP undoes that at its input, so the processing still sees the music's own level. One choice per mode: on by default with one channel, off with two decks (a key on the keyboard should not move a DJ's outputs).
- **Start DJAP when Windows starts** (2.5). Stored in Windows' list of startup programs, not in the settings file.
- **Show tooltips** - on (the default): every control explains itself when the mouse rests on it. Off: no tooltips in any window. The plug-in follows the setting the next time it is loaded.

5.3 Updates

- **Check GitHub for a new version once a day** (among the switches) - on by default.

The rest is in the **About DJAP** window (right-click the main window or the notification-area icon: *About DJAP...*):

- **Check now** asks at once.

- **Install update** downloads and installs a newer version (chapter 11). Greyed when there is nothing newer.
- The text beside the buttons shows this version and what the last check found.

5.4 Start and stop

Right under Look-ahead and Oversampling:

- **Start audio** starts the audio with the choices above (with WASAPI exclusive it asks first).
- **Stop audio** fades out and stops.

5.5 Health

At the bottom of the window, under one heading:

- What runs: the driver or devices, rate, buffer, the latency of input, output and processing, and per deck the extra latency and drift of an input on another clock; a message when the audio could not start or the settings could not be saved.
- Under it, counters that show whether the computer keeps up with the audio (chapter 10). **Reset counters** sets them to 0.

6. Loudness presets and the loudness window

6.1 The shipped loudness presets

A loudness preset holds the target loudness and every value of every stage. Five loudness presets are named by strength, three by the medium they imitate. In the loudness preset lists they appear from the quietest output to the loudest:

Loudness preset	Output	Character
DAB+	-23 LUFS	Digital radio the EBU R 128 way: the music keeps its dynamics; songs are moved to the target slowly instead of squashed; only passages above -18 LUFS are compressed; no density, clipping or exciter. Quiet next to the others: turn the amplifier up.
Gentle	-18 LUFS	Easy listening: songs are brought to one loudness slowly and gently, most of the dynamics stay. Light compression, 3 density bands, no clipping. For background music and quiet rooms.
Light	-16 LUFS	A little more even and present than Gentle, still natural. 4 density bands, the de-esser and a touch of clipping.
Web Stream	-16 LUFS	Internet radio, podcasts, MP3 / AAC files: like Light but without clipping and with the ceiling at -2 dBTP, because lossy coding raises peaks and turns clipping into audible distortion.
Medium	-14 LUFS	The usual loudness of streaming services: even and full without sounding squashed. 5 density bands, a little exciter, the de-esser. A good start for most music.
FM Station	about -13 LUFS	The sound of an FM station: processed like a loud station (fast, dense 5 bands, full low end, slightly wider, hard clipping, highs held down) and then turned down 4 dB, as the transmitter's power limit does. It sounds like FM; it is not a signal ready for a transmitter.
Loud	-12 LUFS	Dense and punchy: earlier and harder compression, 6 density bands, bass enhancer and exciter, more clipping. For parties and noisy rooms.
Very Loud	-10 LUFS	Close to modern club masters: very dense and loud, little dynamics left. 7 density bands, the strongest compression and clipping. For loud venues; on long listening it can tire.

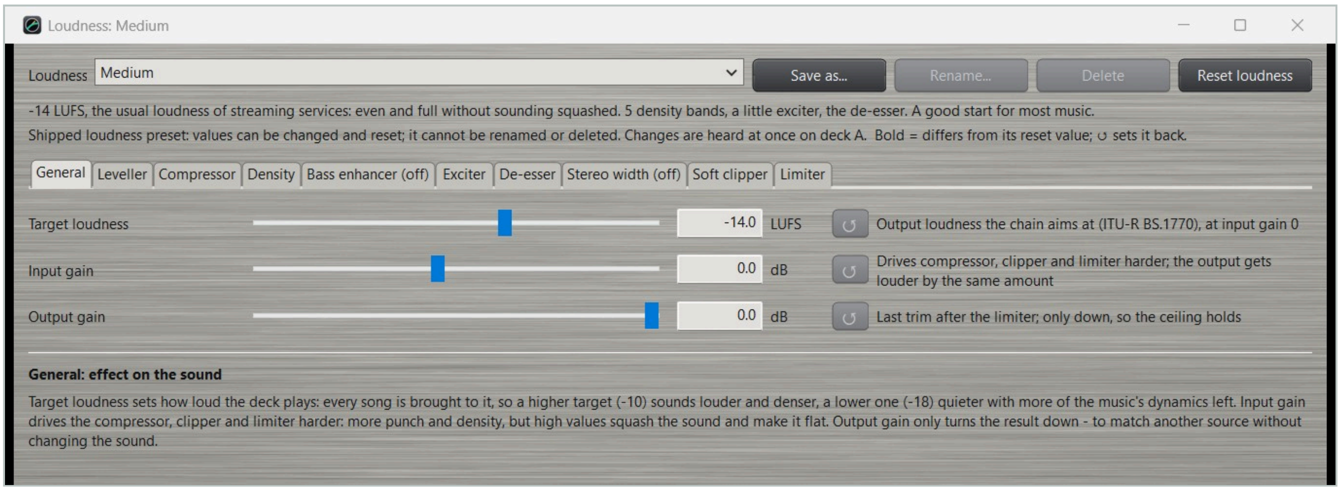
All loudness presets have a limiter ceiling of -1.0 dBTP except Web Stream (-2.0 dBTP).

The main values (`DJAPCmd --presets` lists them for the loudness presets on your computer, including your own):

Loudness preset	Target LUFS	Leveller range / speed	Compressor threshold / ratio	Density bands	Clipper drive	Ceiling dBTP
DAB+	-23	15 dB / 0.5 dB/s	+5 LU / 2.0	off	off	-1.0
Gentle	-18	12 dB / 0.5 dB/s	+6 LU / 1.5	3	off	-1.0
Light	-16	9 dB / 1.0 dB/s	+4 LU / 1.8	4	1 dB	-1.0
Web Stream	-16	9 dB / 1.0 dB/s	+4 LU / 1.8	4	off	-2.0
Medium	-14	9 dB / 1.0 dB/s	+2 LU / 2.0	5	2 dB	-1.0
FM Station	-9 (out -4 dB)	12 dB / 2.0 dB/s	-3 LU / 3.5	5	6 dB	-1.0
Loud	-12	12 dB / 1.5 dB/s	0 LU / 2.5	6	3 dB	-1.0
Very Loud	-10	12 dB / 1.5 dB/s	-2 LU / 3.0	7	4.5 dB	-1.0

These values are starting points; change them to taste.

6.2 The loudness window



Loudness: Medium

Loudness: Medium

-14 LUFS, the usual loudness of streaming services: even and full without sounding squashed. 5 density bands, a little exciter, the de-esser. A good start for most music.

Shipped loudness preset: values can be changed and reset; it cannot be renamed or deleted. Changes are heard at once on deck A. Bold = differs from its reset value; ↺ sets it back.

Target loudness: -14.0 LUFS – Output loudness the chain aims at (ITU-R BS.1770), at input gain 0

Input gain: 0.0 dB – Drives compressor, clipper and limiter harder; the output gets louder by the same amount

Output gain: 0.0 dB – Last trim after the limiter; only down, so the ceiling holds

General: effect on the sound

Target loudness sets how loud the deck plays: every song is brought to it, so a higher target (-10) sounds louder and denser, a lower one (-18) quieter with more of the music's dynamics left. Input gain drives the compressor, clipper and limiter harder: more punch and density, but high values squash the sound and make it flat. Output gain only turns the result down - to match another source without changing the sound.

Click **Edit...** next to a deck's loudness preset. At the top:

- **Loudness** - the loudness preset shown; choosing another one here also moves the deck it was opened from to it.
- **Save as...** saves the current values under a new name; the deck moves to the new loudness preset.
- **Rename...** and **Delete** - for your own loudness presets only. Deleting a loudness preset moves the decks that used it to Medium.
- **Reset loudness** - back to the shipped values (for an own preset: to the general defaults).
- Under it, what the loudness preset does to the sound, and "Some values are changed" when they differ from the shipped ones.

Below that, a **tab per stage**. A stage that is switched off says "(off)" on its tab. On each tab:

- The stage's **On** check box first.
- A row per value: name, **slider**, the **value** (it can be typed - press Enter or leave the field; a comma counts as the decimal sign), the unit, a small **reset** button (back to the reset value; only enabled when the value differs, and its tooltip names the reset value) and a short explanation. A value in **bold** differs from its reset value. The row's tooltip has the full explanation and the range.
- Under the rows, what the stage does to the sound and what more or less of its values sounds like.

The window can be made as large as you like; it cannot be made smaller than the size that shows every row of the largest tab and all texts in full (a narrower window is taller, because its texts take more lines).

Every change is heard at once; a dragged slider is saved when it is let go. Values outside the range are limited to it; text that is not a value gives a warning. Times and frequencies that span a wide range move on a logarithmic scale.

6.3 All values of a loudness preset

Defaults below are those of Medium (the general defaults of a new own preset).

General

Value	Unit	Range	Default	What it does
Target loudness	LUFS	-24 .. -8	-14	The loudness every song is brought to (at input gain 0). Higher sounds louder and denser; lower keeps more dynamics.
Input gain	dB	-15 .. +18	0	Drives the compressor, clipper and limiter harder: more punch and density; the output gets louder by the same amount. High values squash the sound.

Value	Unit	Range	Default	What it does
Output gain	dB	-15 .. 0	0	A last trim after the limiter, only down, so the ceiling holds. To match another source without changing the sound.

Leveller - the slow hand on the volume.

Value	Unit	Range	Default	What it does
On			on	
Range	dB	0 .. 15	9	The largest boost or cut. Larger evens out bigger differences between songs.
Speed	dB/s	0.1 .. 6	1.0	How fast the gain may move. Higher reaches a new song's level sooner but can be heard as the volume drifting within a song.
Attack	s per 10 dB	0.2 .. 20	1.0	How fast the gain comes down when a song comes in clearly too loud (after a quiet intro or a break). Shorter: the loud part is down sooner; longer: a gentler turn of the knob.
Release	s per 10 dB	0.2 .. 20	1.5 (Gentle, DAB+: 5)	How fast the gain goes up when the music is clearly too soft (a soft intro, a quiet song after a loud one), and to a new song after a pause. Shorter: on level sooner, but the rise is easier to hear; longer: a gentler turn of the knob.
Hold below	LU	5 .. 20	10	Holds the gain while the music is this far under the target: fades and pauses are not pumped up. A quiet part that stays this low for the Quiet wait without fading (a soft intro or verse, a breakdown) is levelled like the rest.
Quiet wait	s	0.5 .. 10	1 (Gentle, DAB+: 3)	How long such a quiet part waits before it is levelled. Shorter: a soft intro comes up sooner; longer: the start of a breakdown or a fade is surer to be left alone.

Attack and Release never move slower than Speed. After a pause (a stopped deck) the next song is caught up at the Release speed (up) or the Attack speed (down). The defaults were chosen on 18 songs with soft intros: they bring such an intro to the target about a quarter sooner than before, without a song coming in louder at its loud part.

Compressor - evens out the loud parts within a song.

Value	Unit	Range	Default	What it does
On			on	
Threshold	LU	-10 .. +15	+2	Relative to the target: material louder than this is compressed. Lower = more compression.
Ratio	:1	1 .. 8	2.0	How strongly. Higher is more even and more "in your face"; too much sounds flat and tired.
Knee	dB	0 .. 12	6	Width of the soft knee: wider starts more gently.
Attack	ms	1 .. 100	10	Short catches drum hits (less punch); longer lets them through (more punch).
Release	ms	20 .. 1000	200	Short sounds lively but can pump; long is smooth.
Side-chain high-pass	Hz	off, 5 .. 200	80	Bass below this does not make the compressor work, so a heavy kick does not pull the whole mix down.
Make-up	dB	auto, 0 .. 12	auto	Gain after compression. Auto keeps the target level unchanged.

Density - the multiband stage.

Value	Unit	Range	Default	What it does
On			on	
Bands		1 .. 7	5	Number of frequency bands (1 = wide-band). More bands act more selectively: the bass no longer pushes the voice around.
Amount	%	0 .. 100	40	Strength: from 1:1 at 0 % to 4:1 at 100 %.

Value	Unit	Range	Default	What it does
Down above	dB	0 .. 12	3	A band this far above its own recent level is turned down.
Lift below	dB	0 .. 20	6	A band this far below its recent level is lifted.
Maximum lift	dB	0 .. 12	6	The largest lift of a band.
Attack	ms	1 .. 200	20	Time to turn a band down.
Release	ms	20 .. 2000	300	Time to turn a band back up.
Split 1 .. 6	Hz	20 .. 16000	per band count	The frequencies between the bands. Only the splits of the band count in use are shown; they always rise, at least 1.25x apart (moving one past its neighbour moves the neighbour). A new band count loads its default splits.
Band 1 .. 7 gain	dB	-12 .. +12	0	A fixed gain per band: a multiband tone control.

Default splits per band count: 2 bands 200 Hz; 3: 200 / 2500; 4: 120 / 600 / 4000; 5: 60 / 200 / 2500 / 6000; 6: 60 / 200 / 800 / 2500 / 6000; 7: 60 / 150 / 400 / 1000 / 2500 / 6000 Hz.

Bass enhancer

Value	Unit	Range	Default	What it does
On			off	
Frequency	Hz	60 .. 160	100	Bass below this gets overtones. Higher is more audible but boomier.
Amount	%	0 .. 100	30	Strength of the overtones; too much sounds muddy.

Exciter

Value	Unit	Range	Default	What it does
On			on	
Frequency	Hz	1500 .. 8000	3500	Highs above this are excited. Lower also brightens voices and snares (can turn harsh); higher only adds air.
Amount	%	0 .. 100	15	Strength; too much sounds hissy, especially on old or low-bit-rate recordings.

De-esser

Value	Unit	Range	Default	What it does
On			on	
Frequency	Hz	3000 .. 10000	5500	Where it listens and acts.
Threshold	LU	-30 .. 0	-10	Relative to the target: sibilance louder than this is reduced (4:1). Lower acts more often.
Maximum reduction	dB	0 .. 12	6	The largest reduction. Too much makes voices lisp and sound dull.

Stereo width

Value	Unit	Range	Default	What it does
On			off	
Width	%	0 .. 200	100	100 % = as recorded, 0 % = mono, above 100 % wider (can sound hollow and loses power in mono).

Value	Unit	Range	Default	What it does
Bass width	%	0 .. 100	100	Width of the bass below the split; 0 % = bass in mono.
Bass split	Hz	40 .. 400	150	Where the bass width starts.

Soft clipper

Value	Unit	Range	Default	What it does
On			on	
Drive	dB	0 .. 9	2	How hard peaks are driven into the rounded curve: more loudness and density, less punch; pushed hard it sounds rough.

Limiter

Value	Unit	Range	Default	What it does
On			on	
Ceiling	dBTP	-6 .. 0	-1.0	The highest true-peak level of the output. Lower leaves more room for converters and lossy encoders.
Release	ms	10 .. 500	80	Short recovers fast (louder, can distort the bass); long is cleaner but can duck the sound after a peak.

Switching the limiter off removes the true-peak guarantee; the output is still clamped at full scale. With the soft clipper on, it then holds the ceiling as a sample peak itself and clips about 1.5 dB under it (that keeps the highs clean); between the samples the level can still rise over the ceiling. For a true-peak ceiling keep the limiter on.

6.4 Tips

- Start with **Medium** and compare one step up or down; judge at the same listening volume.
- For mixed old and new music, a **colour** with auto tone (Oldies or Auto) usually does more for the balance than changing the loudness preset.
- If the volume seems to drift within songs, lower the leveller **Speed**; if new songs take too long to settle, raise it (or rely on catch-up after pauses).
- If the sound "breathes" or pumps, lengthen the compressor and density **Release**, or lower the density **Amount**.
- Watch the stage rows: a limiter that reduces more than a few dB all the time means the stages before it push too hard (lower input gain, clipper drive or target).

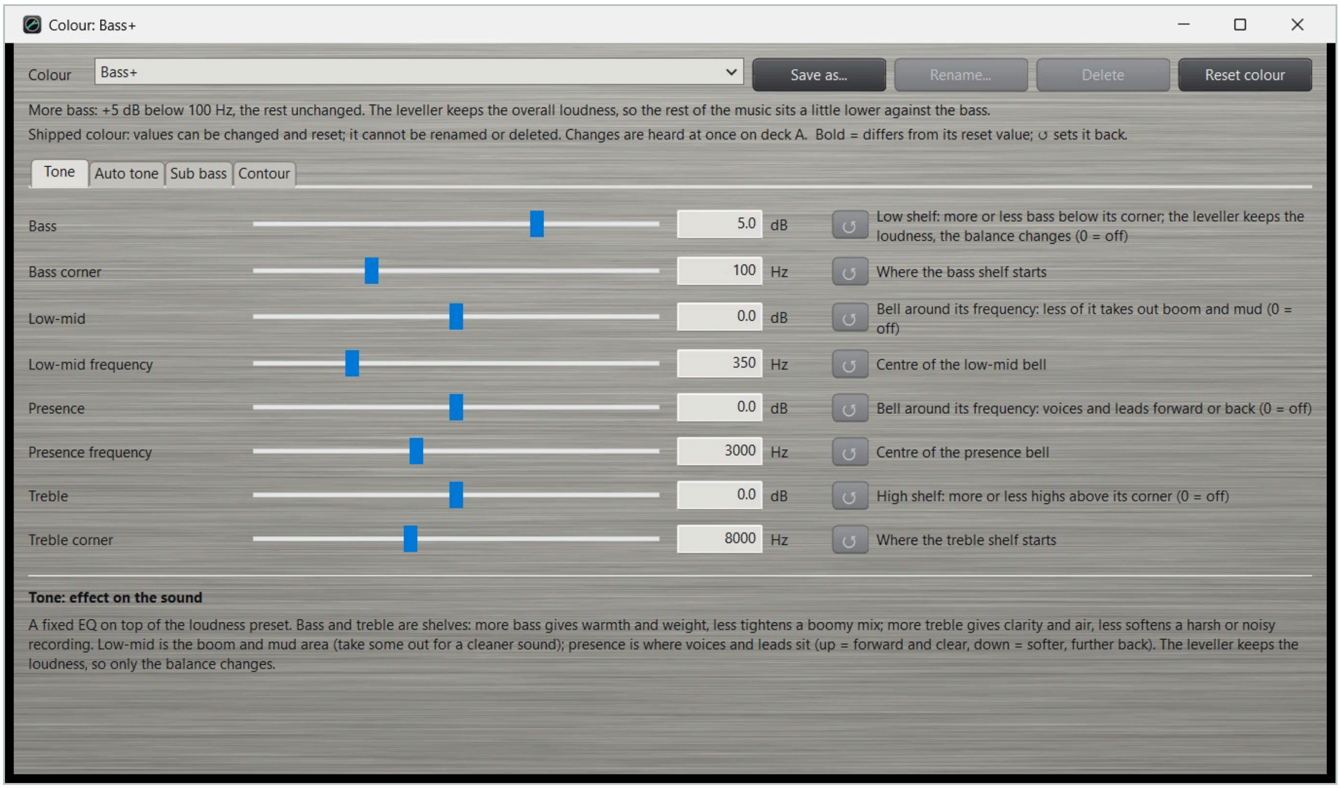
7. Colours and the colour window

7.1 The shipped colours

A colour holds only the tone. The leveller keeps the loudness on target, so a colour changes the balance, not the volume.

Colour	Values	Use
Neutral	everything off	The tone as it is.
Warm	bass +3, low-mid -1, treble -1.5 dB	Fuller and softer; bright or thin mixes, long listening.
Bass+	bass +5 dB	More bass, the rest unchanged.
Less mid	low-mid -3, presence -1.5 dB	Cleaner, less boxy; takes boom and harshness out of live and older recordings.
Bright	presence +2, treble +3 dB	Clearer and more open for dull recordings (sharp on bright ones).
Vocal	bass -1, low-mid -2, presence +2.5 dB	Voices forward: speech, vocals, karaoke.
Oldies	auto tone 70 %, sub bass 20 %, bass +1, treble +1 dB	Thin, flat old recordings of the 60s and 70s.
Auto	auto tone 100 %	Each song pulled towards a well-balanced production, up to 6 dB (treble 3 dB).
Quiet	contour 60 %	Listening at a low volume: deep bass and highs added so the music stays full.

7.2 The colour window



Colour: Bass+

More bass: +5 dB below 100 Hz, the rest unchanged. The leveller keeps the overall loudness, so the rest of the music sits a little lower against the bass.

Shipped colour: values can be changed and reset; it cannot be renamed or deleted. Changes are heard at once on deck A. Bold = differs from its reset value; ↺ sets it back.

Bass	5.0	dB	Low shelf: more or less bass below its corner; the leveller keeps the loudness, the balance changes (0 = off)
Bass corner	100	Hz	Where the bass shelf starts
Low-mid	0.0	dB	Bell around its frequency: less of it takes out boom and mud (0 = off)
Low-mid frequency	350	Hz	Centre of the low-mid bell
Presence	0.0	dB	Bell around its frequency: voices and leads forward or back (0 = off)
Presence frequency	3000	Hz	Centre of the presence bell
Treble	0.0	dB	High shelf: more or less highs above its corner (0 = off)
Treble corner	8000	Hz	Where the treble shelf starts

Tone: effect on the sound

A fixed EQ on top of the loudness preset. Bass and treble are shelves: more bass gives warmth and weight, less tightens a boomy mix; more treble gives clarity and air, less softens a harsh or noisy recording. Low-mid is the boom and mud area (take some out for a cleaner sound); presence is where voices and leads sit (up = forward and clear, down = softer, further back). The leveller keeps the loudness, so only the balance changes.

Click **Edit...** next to a deck's colour. It works like the loudness window (6.2): Save as, Rename, Delete (own colours; deleting moves its decks to Neutral), Reset colour, and a tab per part. The colour window has no on / off switches: a part with all its amounts at 0 is off.

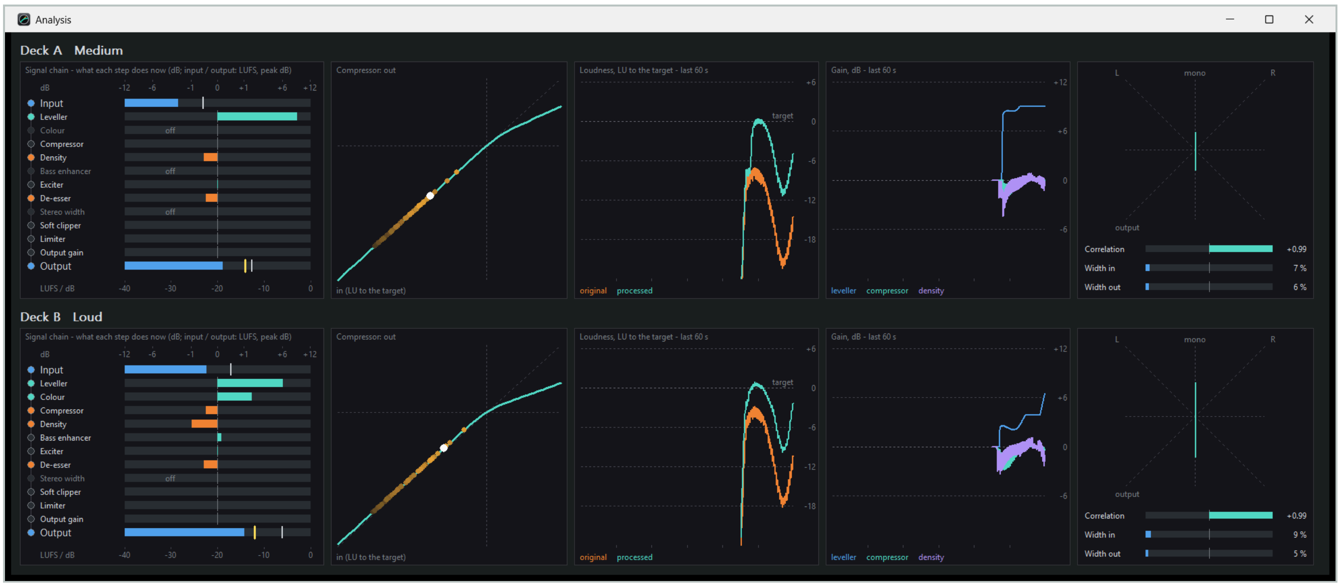
7.3 All colour values

Tab	Value	Unit	Range	Default	What it does
Tone	Bass	dB	-12 .. +12	0	Low shelf: more or less bass below its corner (0 = off).
	Bass corner	Hz	40 .. 250	100	Where the bass shelf starts.
	Low-mid	dB	-12 .. +12	0	Bell: less takes out boom and mud.
	Low-mid frequency	Hz	150 .. 1000	350	Centre of the low-mid bell.
	Presence	dB	-12 .. +12	0	Bell: voices and leads forward or back.
	Presence frequency	Hz	1000 .. 6000	3000	Centre of the presence bell.
	Treble	dB	-12 .. +12	0	High shelf: more or less highs above its corner.
	Treble corner	Hz	3000 .. 16000	8000	Where the treble shelf starts.
Auto tone	Amount	%	0 .. 100	0	How far each song's balance is pulled towards the reference (0 = off).
	Maximum	dB	0 .. 9	6	The largest correction of a band (treble: half of it - the top of an old recording is often its hiss).
	Time	s	2 .. 60	10	How long a band is measured before it is corrected; faster for the first seconds of a song.
Sub bass	Amount	%	0 .. 100	0	Adds a tone an octave below the bass. Use little: it is felt on large speakers and takes headroom.
	Frequency	Hz	40 .. 120	80	Bass below this is followed an octave lower.
Contour	Amount	%	0 .. 100	0	Equal-loudness contour for quiet listening: up to +9 dB deep bass and +4 dB highs. Leave it off at normal volume.

The auto tone holds still during breakdowns and fades, so a quiet part does not change the sound.

8. The analysis and spectrum windows

8.1 The analysis window

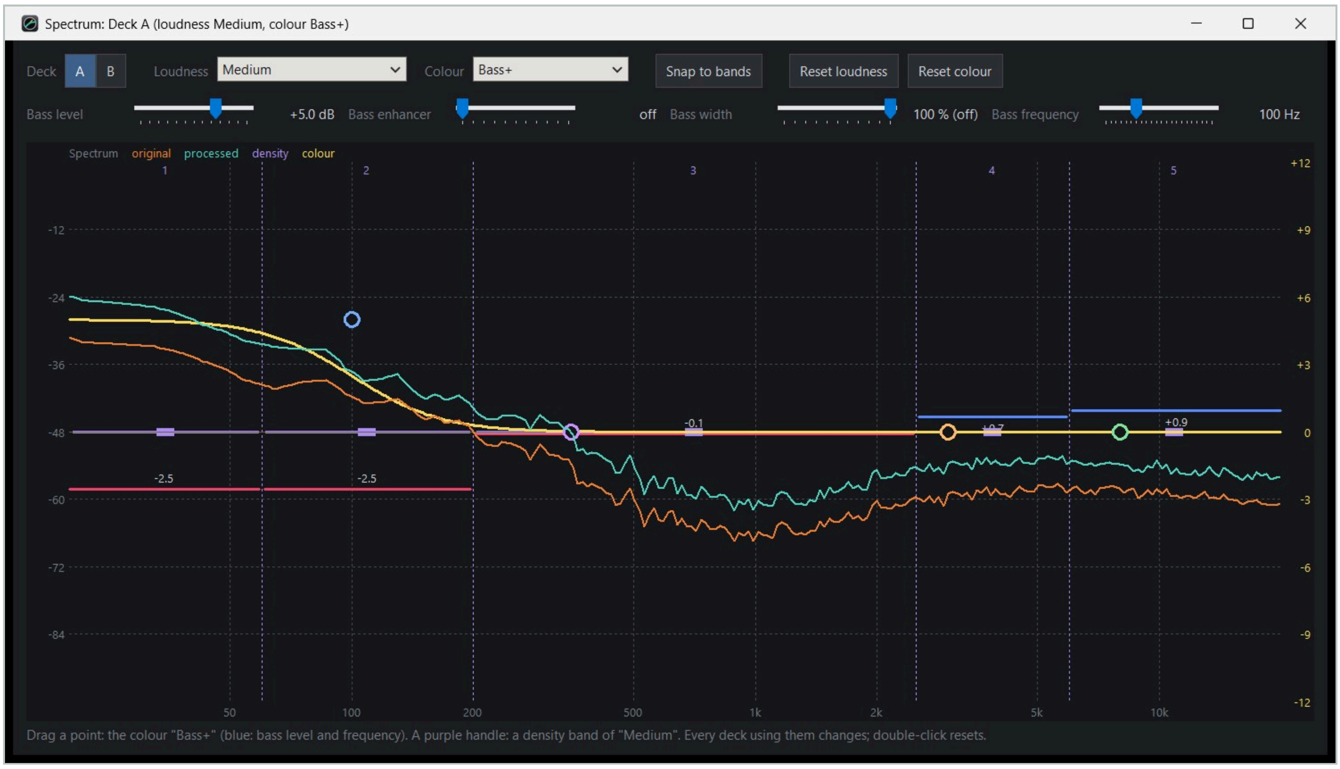

Analysis
Deck A Medium
Signal chain - what each step does now (dB; input / output: LUFS, peak dB)
Input
Leveller
Colour
Compressor
Density
Bass enhancer
Exciter
De-esser
Stereo width
Soft clipper
Limiter
Output gain
Output
off
LUFS / dB
Compressor: out
in (LU to the target)
Loudness, LU to the target - last 60 s
target
original
processed
Gain, dB - last 60 s
leveller
compressor
density
L
mono
R
output
Correlation +0.99
Width in 7 %
Width out 6 %
Deck B Loud
Width in 9 %
Width out 5 %

Click **Graphs** on a deck strip. The window has a row per deck with five panels:

1. **Signal chain** - a row per step in chain order, from *Input* to *Output*. Each step has a lamp (lit when the step changes the level by 0.1 dB or more; turquoise = up, orange = down) and a bar against a dB scale from -12 to +12 (a square-root scale, so small changes are visible). The compressor, de-esser, clipper and limiter show their own gain reduction; the other steps show the level after the step against the level before it. *Input* and *Output* show the momentary loudness as a bar (-40 .. 0 LUFS) and the peak as a white mark; the output's target is the yellow mark. A step that is switched off says "off". **Click a row** to open that step's settings (the loudness window on its tab, or the colour window).
2. **Compressor** - the compressor's static curve for the loudness preset: output against input, in LU relative to the target (the dashed line is 1:1). The white dot is where the detector is now; amber dots show the last 3 seconds.
3. **Loudness** - the momentary loudness of the input (original, orange) and output (processed, turquoise) over the last 60 seconds, relative to the target.
4. **Gain** - the leveller, compressor and density gains over the last 60 seconds.
5. **Scope** - an oscilloscope of the waveform, left channel above and right below, to see whether the sound clips: 50 ms across by default (a tone stands still: the picture starts at a rising zero crossing), or 1, 5, 10, 20 or 200 ms, or the last 2 s (then each moment's highest and lowest sample). The red dotted lines are full scale (0 dBFS); red parts stood at full scale - that is clipping. Under the scope: each channel's peak and the clipped samples of the last 2 s, whatever the time across. For the output the yellow dotted lines are the limiter's ceiling, which the sound should stay inside. **Right-click** the scope for the *input* (orange: what comes in - clipping there is in the source) or the *output* (turquoise), and for the time across.
6. **Stereo** - a goniometer of the output (mono is the vertical line, left and right the diagonals), with bars for the output's correlation (+1 = mono, 0 = unrelated, negative = out of phase) and the stereo width of input and output (0 % = mono). They do not depend on the level: two decks with the same music show nearly the same values whatever their loudness presets. Only the Stereo width stage changes the width on purpose, and at Width 100 % (the default, "as recorded") it leaves it as it is - Width out then lies a percent or so under Width in, from the dynamics.

The history is cleared at every start of the audio.

8.2 The spectrum window



Loudness: Medium

Colour: Bass+

Bass level: +5.0 dB

Bass enhancer: off

Bass width: 100 % (off)

Bass frequency: 100 Hz

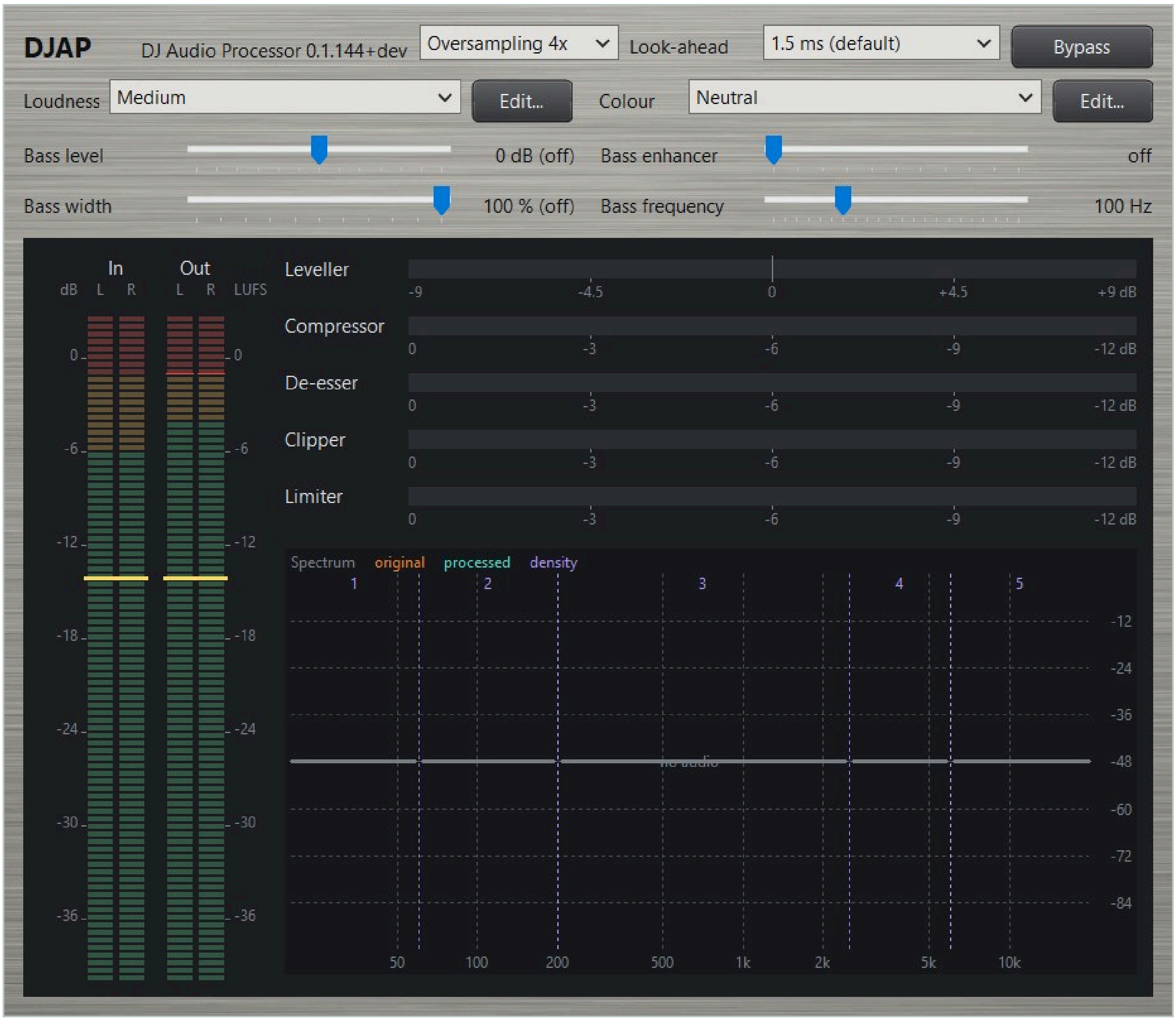
Drag a point: the colour "Bass+" (blue: bass level and frequency). A purple handle: a density band of "Medium". Every deck using them changes; double-click resets.

Click the **spectrum** in a deck's meter panel. The big graph shows, for one deck:

- The **spectrum** of the input (orange) and the output (turquoise), with its dB scale on the left. It reaches down to -96 dB; what is quieter is not drawn, so where a song has (almost) no highs the line drops out at the bottom.
- The **density bands** and their gains now: a purple line per band at its gain on the right-hand scale (above the 0 dB line = lifted, under it = turned down), with the gain in dB.
- The deck's **colour as an EQ curve** (yellow, the right-hand scale in dB): thick - what the colour does now, including the auto tone's correction for this song; dashed - the same without the auto tone (only while it corrects); thin - each tone section on its own. A shaded block marks the sub bass's range.
- A **point per tone section**: bass (blue), low-mid (purple), presence (orange), treble (green). **Drag** a point: left and right change its frequency, up and down its gain, in the same steps as the colour window's sliders. **Double-click** a point to set its gain back to 0 dB. This changes the colour itself - every deck using it
    - is heard at once and saved when you let go of the mouse. The colour window and the bass sliders follow.
- A **purple handle per density band**: the band's fixed gain (*Band N gain* in the loudness preset's Density tab, -12 to +12 dB, right-hand scale), lying exactly over the band. **Drag** it up or down; **double-click** sets it back to 0 dB. This changes the deck's loudness preset (every deck using it). So the density bands work as an EQ of their own, next to the colour's four tone sections.
- **Reset loudness** and **Reset colour** set every value of the deck's loudness preset or colour back, as the buttons in the loudness and colour windows do: a shipped one to its shipped values, your own to the defaults (asks first).
- **Snap to bands** (lit = on, remembered): a colour point dragged near a band split sits exactly on it (the split lights up), so a tone section can start where a density band does. A split outside a section's range is skipped.
- With two decks, **Deck A / B** switches between them. **Loudness** and **Colour** choose the deck's loudness preset and colour, as the lists in the main window do (both decks when they are linked).
- The **four bass sliders** (4.2) - bass level, enhancer, width and frequency - in a second row, for the deck shown, and the curve shows the change at once.
- The bass level and frequency **in the graph**: the blue bass point is the bass level (drag up or down) and the bass frequency (drag left or right; as the slider, it moves the enhancer's and the width's frequency too). The sliders at the top follow.

- The left scale (spectrum, dB) and the right one (colour, density, -12 to +12 dB) share the same grid lines.
- At the top right of the graph the window shows what is under the mouse: the frequency with the input, output and colour levels there, or the point's frequency and gain.

9. The plug-in for DJ software



DJAP DJ Audio Processor 0.1.144+dev

Oversampling 4x

Look-ahead 1.5 ms (default)

Loudness Medium

Colour Neutral

Bass level 0 dB (off)

Bass enhancer off

Bass width 100 % (off)

Bass frequency 100 Hz

Bypass
Edit...
Edit...
In
Out
dB L R
L R LUFS
0
-6
-12
-18
-24
-30
-36
Leveller
-9 -4.5 0 +4.5 +9 dB
Compressor
0 -3 -6 -9 -12 dB
De-esser
0 -3 -6 -9 -12 dB
Clipper
0 -3 -6 -9 -12 dB
Limiter
0 -3 -6 -9 -12 dB
Spectrum original processed density
1 2 3 4 5
no audio
-12
-24
-36
-48
-60
-72
-84
50 100 200 500 1k 2k 5k 10k

The plug-in brings DJAP's processing into DJ software (or any other program that loads plug-ins): one deck per plug-in, stereo in and stereo out, with exactly the same chain, loudness presets and colours as the program. The DJ software keeps the audio interface; DJAP only processes what passes through it.

9.1 Installing and finding it

Choose "Plug-in for DJ software" on the setup's "Select Components" page (it is chosen by default). It needs an installation for all users (the normal one): the plug-in goes into the computer's plug-in folder, `C:\Program Files\Common Files\VST3\DJAP\DJAP.vst3` (one file in a folder of its own), where DJ software looks for plug-ins. If the DJ software does not list "DJAP" yet, let it scan its plug-ins again (its settings usually have a button for that) and restart it.

Put one plug-in on each deck (as an effect of that deck or channel), or one on the master. The plug-in tells the DJ software its delay (the look-ahead, 1.5 ms by default), so the DJ software can keep the decks in time.

The program and the plug-in can be installed and used together or apart.

9.2 The window

The window is one deck strip of the program:

- **Oversampling** - 2x, 4x (default) or 8x, as in the Setup window (5.1); stored with the DJ software's project.
- **Look-ahead** - the limiter's look-ahead, 0, 1.5, 2, 3 or 5 ms (see 5.1), a list of these five in the DJ software too; the DJ software is told the new delay. An old project's 0.5 or 1.0 ms runs as 1.5 ms.
- **Bypass** - lit: the input goes out unchanged, delayed by the same look-ahead.
- **Loudness** and **Colour** - the lists of the program (see 9.3), each with its **Edit...** button: the same loudness preset and colour windows as in the program (chapters 6 and 7), with every value, Save as, Rename, Delete and Reset.
- **The four bass sliders** (4.2).
- **The meter panel** - LED bars, the gain of each stage and the spectrum with the density bands and the colour curve, as in the main window (4.3). Clicking a stage's row or a column of bars opens the loudness window on that stage's tab, as in the program; clicking the spectrum opens its Density tab.

9.3 Loudness presets, colours and values

- The loudness preset and colour lists come from the program's settings file (`%APPDATA%\DJAP\DJAP.ini`): the shipped loudness presets and colours, and your own. Without the program the plug-in has the shipped ones, and works just the same.
- Loudness presets and colours made or changed in the plug-in's loudness window are saved in that same file, so the program and every plug-in share them. Program and plug-ins never overwrite each other's loudness presets: each saves only its own changes into the file as it is at that moment. The plug-in never changes the program's other settings. A loudness preset made in the program appears in the plug-in when its loudness window is opened (or when it is loaded).
- Choosing a loudness preset or colour loads its values. Choosing the one that is already chosen loads its values again - back to the loudness preset after changes. "(other values)" means the values match no loudness preset in the list (for example a project made on another computer).
- **Every value is a parameter** the DJ software can show and automate: the 72 values of the loudness preset and colour (named "Stage: value", for example "Compressor: Threshold", grouped by stage), the loudness preset and colour lists, Bypass and the look-ahead. They move in the same steps as the program's sliders.
- A **project** (or a preset of the DJ software) stores the values themselves, not only the loudness preset's name: it sounds the same on a computer where that loudness preset does not exist or was changed.

9.4 Updates

The plug-in is updated by the same setup as the program (chapter 11). If the DJ software has the plug-in loaded during an update, the setup never closes the DJ software: the new plug-in takes the old one's place the next time Windows starts.

10. Latency and real-time health

Latency is the time from the input socket to the output socket. It has three parts:

- the device's input buffer and output buffer (set by the buffer size; ASIO reports them exactly),
- the processing: only the limiter look-ahead (1.5 ms by default; 66 samples at 44.1 kHz) - the attack of every hit comes out after that, plus a fraction of a millisecond. The filters that split the spectrum let the lowest frequencies follow a few milliseconds later (Medium: about 6 ms at 100 Hz, 11 ms at 50 Hz; the technical manual, 1.5, lists each loudness preset), as in any multiband processor; both decks on the same loudness preset stay aligned.
- with an input on another clock (an ASIO output fed from a WASAPI / DirectSound / MME input, or WASAPI / DirectSound / MME with Drift correction): the drift buffer - the input's block plus the output's period plus 3 ms, so about 8 ms at 128 frames and 23 ms at 10 ms periods. After a dropout of the input it grows by 2 ms (up to 20 ms more) and shrinks again by 0.5 ms per quiet minute; the burst of sound a device sends after a hiccup can raise it for 20-40 seconds, and it then comes down slowly (0.3 ms a second, a speed change too small to hear) - after a long hiccup (100 ms) the buffer is back at its usual size after about 3 minutes. A device that hiccups again and again for 70 ms or more gives a dropout each time. With WASAPI Windows may run longer periods than the buffer size asked for (its low-latency mode takes only some sizes, at the device's own rate): the Setup window then says so, and the latency shown follows.

The Setup window shows the total.

Health (Setup window) shows whether the computer keeps up:

Counter	Meaning
Periods	Audio periods processed since the start (or the last reset).
Heaviest / last	The processing time of the heaviest and the last period, in % of the period's length.
Over half their time	Periods that needed more than half of their time - a warning sign.
Late (dropout)	Periods not ready in time: you hear a dropout.
Answered with silence	The engine was not ready (for example while starting).
Driver resyncs, overloads	Reported by the ASIO driver.

If late periods appear, raise the buffer size, close other heavy programs, or use fewer density bands. The processing of both decks together uses only a few percent of one CPU core.

11. Updates

With "Check GitHub for a new version once a day" on (default), DJAP asks the project's public GitHub page about 15 seconds after it starts and then once a day. It only asks; it sends nothing about you or your computer beyond what any web page request contains. GitHub answers its quick way of asking only 60 times an hour per network (every program behind your router counts); when that is used up DJAP reads the release page itself instead, which GitHub does not limit.

When a newer version exists, DJAP tells you once with a message at its notification-area icon, in the status bar and in the About window. **Nothing is downloaded or installed until you click Install update** in the About window (clicking the message opens it).

Install update:

1. asks first while the audio runs: the audio stops during the update (usually less than half a minute) and starts again by itself with the new version;
2. downloads the new setup and checks it against the SHA-256 checksum GitHub publishes for it - a setup that does not match, or has no checksum, is never run;
3. closes DJAP (with a fade-out) and runs the setup silently for the same folder; Windows may ask for permission;
4. starts DJAP again with your settings - and the audio, when it ran before. "Start with Windows" stays as you set it.

If something goes wrong the Setup window says what. The setup's log of the last update is in `%TEMP%\DJAP-update\setup.log`.

12. The command-line program DJAPCmd

`DJAPCmd.exe` (in the installation folder) processes files offline with exactly the same code as the live program. It never plays audio and never changes the settings. Open a command prompt in the installation folder, or give its full path.

```
DJAPCmd --render <in> <out.wav> [--a <preset|off>] [--b <preset|off>] [--ca <colour>] [--cb <colour>]
[--look <ms>] [--oversampling <2|4|8>] [--keep-delay] [--pcm24] [--trace <file.csv>]
DJAPCmd --presets
DJAPCmd --measure <file.wav> [more files]
DJAPCmd --tone <file> [more files]
DJAPCmd --devices
DJAPCmd --ini <file> ...     (use another settings file, read only)
DJAPCmd --version | --help
```

- **--render** processes a file. Input: WAV, or MP3 / M4A / WMA / FLAC through Windows Media Foundation (mono plays on both channels). A 2-channel file is deck A; a 4-channel file is deck A (channels 1-2) and deck B (3-4). Each deck uses its loudness preset and colour from the settings unless `--a` / `--b` (loudness preset, or `off` for unprocessed) and `--ca` / `--cb` (colour) say otherwise. The output is 32-bit float WAV (or 24-bit PCM with `--pcm24`), with the look-ahead delay removed (kept with `--keep-delay`: the file is then longer by the delay, so the look-ahead cuts nothing off the end - the filters' last few milliseconds of ringing after an abrupt end are still left out, about -35 dB); over 4 GB (about three hours of 4-channel 24-bit audio) it is written as RF64, the 64-bit form of WAV. The file is written under a temporary name first, so a failed render leaves no half file. A render holds the whole input and output in memory (about 3 times the size of a 24-bit file): fine for songs and sets, but a file of several GB needs that much free memory. `--oversampling` sets the clipper's and exciter's oversampling for this render (default: the settings'). After rendering it prints a report of input and output loudness. `--trace <file.csv>` also writes deck A every 100 ms: loudness in and out (momentary and short-term), the leveller's gain and whether it holds, and what the compressor, density bands, de-esser, clipper and limiter do - to see why a song comes out uneven. The trace file must be a file of its own (not the input or the output).
- **--presets** lists all loudness presets and colours with their main values and which deck uses them.
- **--measure** prints the integrated loudness, loudness range, maximum momentary and short-term loudness, sample peak and true peak of WAV files (per deck for 4-channel files).
- **--tone** prints each file's long-term tone balance (60 Hz, 250 Hz, 3 kHz and 10 kHz against 1 kHz) and what the auto tone would correct at 100 %.
- **--devices** lists the ASIO drivers and the WASAPI / DirectSound / MME devices without opening any of them.

Example - process a song with the Loud loudness preset and the Oldies colour into a 24-bit file:

```
DJAPCmd --render song.mp3 song-loud.wav --a Loud --ca Oldies --pcm24
```

Exit code: 0 done, 1 wrong command line, 2 a file could not be read or written.

13. The settings file

All settings, the decks' choices and your own presets and colours are kept in one text file, `DJAP.ini`:

- next to the program when that folder is writable (a portable copy),
- otherwise (an installation in Program Files) in `%APPDATA%\DJAP\DJAP.ini`.

The About window shows which file is in use. DJAP saves it after every change. It can be read and edited by hand - **only while DJAP is closed**. Sections:

Section	Holds
`[Audio]`	Processing (`Decks`), API, driver, devices, channel pairs, rate, buffer, look-ahead, `StartAudio` (start the audio at program start), follow the Windows volume
`[Decks]`	Each deck's loudness preset, colour and processing switch, linked presets
`[Window]`	Where each window was last - position, size, maximised (a window opens there; without a line it opens in the middle of the screen) - and the window options
`[Updates]`	`CheckUpdates` (1 = the daily check is on)
`[Preset <name>]`	The values of a loudness preset that differ from its shipped values (own loudness presets: all values)
`[Colour <name>]`	The same for a colour

To start over, close DJAP and delete (or rename) the file; DJAP starts with the shipped loudness presets and opens the Setup window. Back the file up to keep your own loudness presets.

Settings of versions up to 0.1.57 (which had another program name) are copied once at the first start of DJAP; the old file stays.

14. Troubleshooting

Problem	What to do
No sound	Is the audio started (Setup window)? Are the right input and output pairs or devices chosen? Is a deck on *(none)*? Do the In bars move? If In moves and Out does not, check the output route and the amplifier.
"Open driver" fails	Another program (DJ software) may hold the ASIO driver. Close it or use another interface. Check that the interface is connected and its driver installed.
A device shows "(not found)"	The saved device is not connected, or it was chosen under another API (each API names devices differently). Choose it again.
"The settings file could not be read" at the start	Another program held `DJAP.ini` (a virus scanner, a sync tool) for more than two seconds. DJAP runs with its default settings and saves nothing, so your file stays as it is; close that program and start DJAP again.
Dropouts, crackles	Look at Health: late periods? Raise the buffer size; close heavy programs; set Windows' power plan to high performance; use fewer density bands. With an input on another clock, underruns in the Setup window mean the input fell behind: its buffer grows by itself, and shrinks again after a few calm minutes.
The volume keys do nothing	Switch on "The outputs follow the Windows volume" in the Setup window.
The sound pumps or breathes	Lower the density amount or the compressor ratio, lengthen their release, or choose a gentler loudness preset.
A loud song starts too loud after a quiet intro	That is the leveller's 3 s memory; "Attack" (Leveller tab) brings it down within about a second. Shorten it if needed - or lengthen Release or Quiet wait, so the soft intro is turned up less.
A soft intro takes too long to come up	Shorten "Quiet wait" (it starts sooner) and "Release" (it rises faster). A very soft intro (more than Range under the target) cannot be brought up fully: raise Range.
Quiet passages come up too much	A quiet part that stays is levelled after a few seconds. To keep more of the difference, lower the leveller range or its speed.
A soft verse after a loud chorus stays too soft	The leveller waits until the music has stopped falling (about 2-4 s) and then, when it is more than 4 dB too soft, comes up at the "Release" speed until it is within 1 dB. A shorter Release brings it up sooner, a larger Range further.
Everything sounds too dull / too sharp	Try another colour (Bright, Warm), or lower the exciter / de-esser.
Limiter works hard all the time	Lower the input gain, the clipper drive or the target, or use a quieter loudness preset.
DJAP does not start with Windows	Check the option in the Setup window and Windows' startup apps list.
The update failed	The Setup window says why; `%TEMP%\DJAP-update\setup.log` has the setup's log. The setup can always be downloaded from the GitHub page and run by hand.
The DJ software does not list the plug-in	Was "Plug-in for DJ software" chosen in the setup, in an installation for all users? Is `C:\Program Files\Common Files\VST3\DJAP\DJAP.vst3` there? Let the DJ software scan its plug-ins again and restart it. The DJ software must be 64-bit.
The plug-in's loudness preset list lacks my own loudness presets	The plug-in reads the program's settings in `%APPDATA%\DJAP\DJAP.ini` when it is loaded and when its loudness window is opened: click Edit... once, or load the plug-in again. A portable program (settings next to `DJAP.exe`) is not seen.
DJAP closed unexpectedly	Start `DJAP.exe --diag <file.txt>` from a command prompt; when it happens again, the file holds a report that helps to find the cause.

15. Glossary

Term	Meaning
LUFS	Loudness Units relative to Full Scale: loudness measured as the ear hears it (ITU-R BS.1770). -14 LUFS is louder than -18 LUFS by 4 LU.
LU	Loudness Unit: a difference in loudness; 1 LU = 1 dB.
Momentary / short-term / integrated loudness	Loudness over 0.4 s, over 3 s, and over a whole programme.
dBFS	Level relative to digital full scale (0 dBFS = the largest sample value).
True peak, dBTP	The peak of the waveform between the samples, as a converter or encoder rebuilds it; it can be higher than the highest sample.
Ceiling	The highest level the limiter lets through.
Look-ahead	The limiter delays the audio slightly so it sees a peak coming and can turn down smoothly before it arrives.
Threshold, ratio, knee, attack, release	Where a compressor starts, how strongly it works, how gradually it starts, how fast it reacts and how fast it lets go.
Multiband	Processing the audio in separate frequency bands.
Bypass	The deck's input goes out unprocessed.
ASIO	A low-latency audio driver standard for audio interfaces.
WASAPI	Windows' own audio system; *exclusive* mode takes a device for one program.
Pre-fader	A signal taken before the mixer's channel fader, so fader moves do not change it.
Goniometer	A stereo display: mono is a vertical line, wide stereo a cloud.
Correlation	How similar left and right are: +1 mono, 0 unrelated, -1 out of phase.

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